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Course: Calculus Beta

Assignment: MyLab Math - Uge 3

Evaluate the integral using integration by parts.

$$\int x e^{11x} dx$$

The general formula for integration by parts is shown below.

$$\int u dv = uv - \int v du$$

If $\int u dv = \int x e^{11x} dx$, determine good choices for u and dv .

Choosing $dv = x dx$ leads to the antiderivative becoming a quadratic polynomial. This would not simplify the integration.

Therefore, let $u = x$ and $dv = e^{11x} dx$ and integrate by parts.

If $u = x$, then $du = 1 dx$. If $dv = e^{11x} dx$, determine v .

$$\begin{aligned} v &= \int e^{11x} dx \\ &= \frac{1}{11} \int 11 e^{11x} dx \\ &= \frac{1}{11} e^{11x} \end{aligned}$$

Now, calculate uv . Substitute and multiply.

$$\begin{aligned} uv &= x \left(\frac{1}{11} e^{11x} \right) \\ &= \frac{1}{11} x e^{11x} \end{aligned}$$

Next, calculate $v du$. Substitute and multiply.

$$\begin{aligned} v du &= \frac{1}{11} e^{11x} dx \\ &= \frac{1}{11} e^{11x} dx \end{aligned}$$

Substitute the expressions in the formula and integrate.

$$\begin{aligned} \int u dv &= uv - \int v du \\ &= \frac{1}{11} x e^{11x} - \int \frac{1}{11} e^{11x} dx \\ &= \frac{1}{11} x e^{11x} - \frac{1}{121} \int 11 e^{11x} dx \\ &= \frac{1}{11} x e^{11x} - \frac{1}{121} e^{11x} + C. \end{aligned}$$

$$\text{Therefore, } \int x e^{11x} dx = \frac{1}{11} x e^{11x} - \frac{1}{121} e^{11x} + C.$$